

What Do We Already Know? Towards Mindful AI for Climate Change Policymaking

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Abstract

The rapid adoption of AI technologies presents a paradox in climate change policymaking: while AI holds transformative potential for advancing climate action such as optimising renewable energy systems, enhancing carbon modelling, and supporting climate adaptation strategies, it also incurs significant environmental costs. Inspired by a discussion that took place as part of an ongoing co-design study involving interdisciplinary stakeholders, this paper aims to provoke dialogue and critical reflection among HCI scholars on the role of AI in the development of climate change policy. Rather than providing conclusive answers, this paper puts forward a research agenda around the uncertainties, trade-offs, and ethical dilemmas that must be navigated as AI becomes increasingly integrated into climate change policy development.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*.

Keywords

AI; Climate Change; Policymaking; SHCI

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1 Introduction

Vignette. During a recent workshop centred on the co-design of services to make complex near-real-time greenhouse gas (GHG) emissions data accessible and actionable, a diverse group of policymakers, scientists, city officials and service designers gathered to evaluate a storyboard prototype for an AI-driven tool. The proposed tool would help predict the impacts of climate policy: enter two policy ideas and forecast changes in GHG emissions and related metrics such

as liveability or local air quality. The conversation initially focused on how an AI tool such as this could provide useful data-based insights, but a city official raised a pivotal question: 'What do we already know?' She went on to argue that before using AI for predictions, city officials and policymakers should assess the data and insights they already have, understand what worked or didn't work in the past, and learn from the actions of similar cities.

The vignette above captures a conversation that took place during a workshop carried out as part of an ongoing research project investigating the co-design of GHG monitoring services to support cities in pursuing carbon neutrality. Although the workshop focused on a specific service concept, this experience led us to reflect more broadly on the role of AI in climate policy development, acknowledging its potential to provide valuable climate insights, but also concerns about its limitations. For example, the risks of overlooking existing knowledge and the environmental cost of deploying AI systems were central considerations.

Accounting for technology's unintended consequences is a prominent topic within the HCI research community with past work identifying AI as both a contributor to and a potential solution for climate change [10, 12, 26, 42]. Focusing specifically on the role of AI in climate change policy-making, this paper poses the central question: How do we as HCI researchers help to reconcile the environmental costs of AI with its potential support efforts towards climate change mitigation?

Discourses around technology and policy design are typically held separately, speaking to different audiences and fields of expertise. However, several recent HCI and CSCW studies have delved into their inherent and required entanglements. For example, research has highlighted how digital platforms intersect with environmental policies and legal structures [11], strategies for bridging design-led approaches with policy evidence requirements [47], and opportunities for HCI practitioners to design technology and its policies simultaneously [51].

To expand this work, in this paper, we put forward a research agenda to investigate the opportunities and tensions surrounding the use of AI in climate change policymaking, highlighting accessibility to climate data, supporting practical action, citizen participation, and considerations around ownership, responsibility, and regulation as critical points of focus.

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2 AI in the Context of Climate Change Policymaking

The workshop described in the vignette was carried out in the context of an ongoing European Union Horizon 2020 project on developing scientific knowledge and capacity for systematic, real-time observations of greenhouse gas emissions in urban areas. Contributing to this project, our work is specifically focused on collaborating with diverse city actors such as policymakers, city officials, scientists, and citizens to design services that can make these valuable data accessible and usable to city actors outside of the scientific research domain. In this paper, this co-design work is deliberately positioned as a starting point to open up novel dialogues rather than a conclusive research output.

Data-Driven Policy Prediction Service

The prototype of the data-driven policy prediction service described in the vignette was created on the foundation of five prior co-design workshops with diverse city actors across four European cities. During these workshops, we focused on exploring potential ways to “translate” specialised atmospheric data on GHG emissions into actionable insights for urban policymakers. The outcomes included the identification of key actors, the definition of service goals, and early ideas for novel digital tools for supporting data-informed climate actions.

The policy scenario prediction service is one of three prototypes that were developed as a manifestation [32, 52] of the design ideas created by workshop participants. The storyboard format enabled us to present the concept of the tool while also inviting participants to add to the story by including empty frames that were either blank or asked open questions (see the snippet from the storyboard below). As described in the vignette, engagement with the prototype surfaces critical tensions and emerging opportunities that can guide mindful AI development (i.e., designed with awareness of its environmental impact) among participants. While this work is ongoing, its value lies in its ability to surface unresolved tensions and spark debate about the future of AI design in climate policy. The tensions emerging from our study draw attention to complex trade-offs associated with using AI to support emissions reduction.

3 Tensions in AI and Climate Change Mitigation

Past HCI work has explored the intersections between climate action and artificial intelligence, unpacking the tensions that exist between the opportunities to support climate change mitigation and pressing concerns about the unintended consequences of AI use in this domain [10, 12, 26, 29, 30, 42]. To give an overview of these studies, we synthesise the findings across four themes: efficiency and accuracy, environmental costs, creative engagement and social complexities.

Efficiency and Accuracy. Research has demonstrated that machine learning (ML) models can significantly contribute to a more sustainable and resilient energy infrastructure. By enhancing energy efficiency, optimising grid operations, and improving renewable energy forecasting [12, 26, 42], ML helps to address critical challenges in the energy sector. Moreover, artificial intelligence can

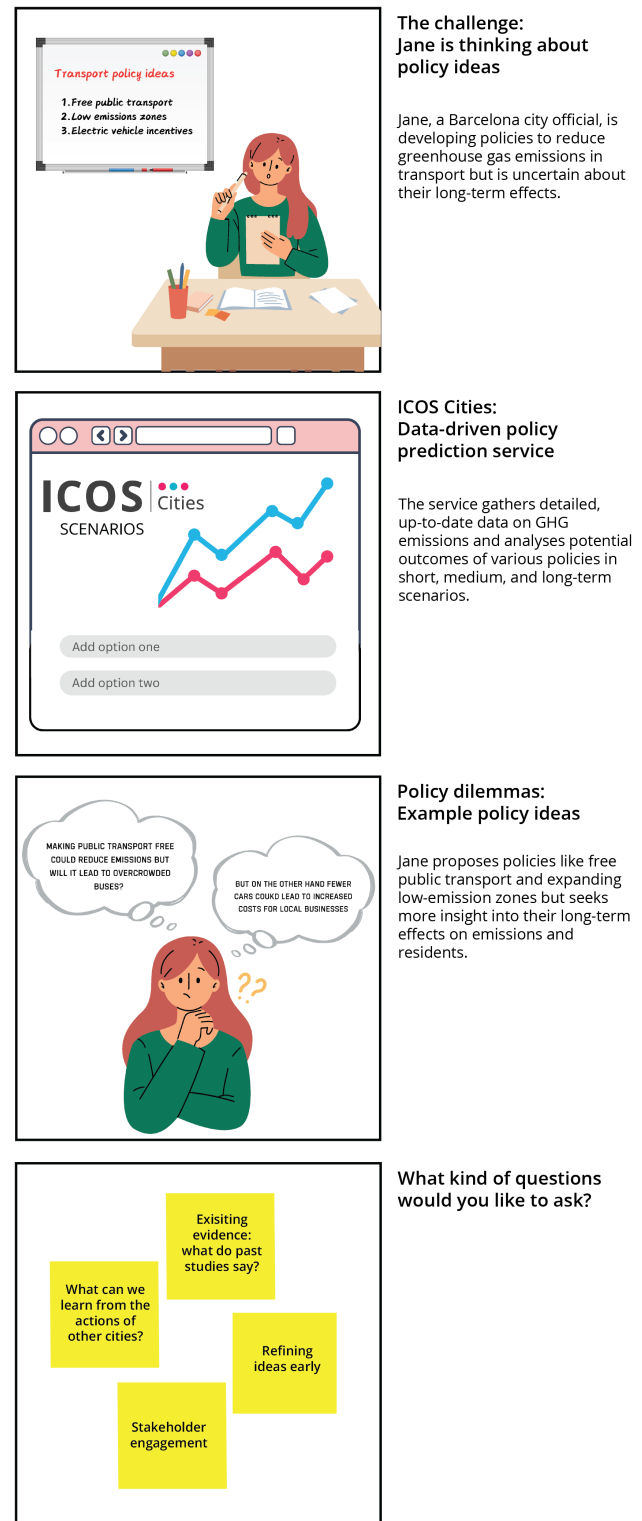


Figure 1: Snippet of the storyboard prototype of the data-driven policy prediction service evaluated in the co-design workshop

help to refine climate models, leading to more accurate predictions and a deeper understanding of climate dynamics. This advancement is vital for informed decision-making and proactive planning in response to climate challenges [26].

Environmental Costs. However, the same authors [12, 26] have also drawn attention to the significant energy and carbon costs associated with training large AI models, warning that, if deployed without careful consideration, AI could unintentionally exacerbate emissions—for example, by improving the efficiency of fossil fuel extraction. For instance, it has been estimated that a single full training cycle of large language models (LLMs) like ChatGPT generates approximately 300,000 kg of CO₂, with each query emitting around 4.32 grams of CO₂ [13]. In addition to this, past research has drawn attention to issues of water usage for cooling data centres, e-waste, and depletion of rare earth minerals connected to the development and use of AI technologies [18, 48].

Creative Engagement. Beyond technical applications, research into AI and climate change mitigation also offers innovative approaches to public engagement by integrating AI-generated visuals and narratives that transcend traditional methods of persuasion [29, 30]. These tools enable the creation of alternative perspectives, fostering emotional connections and engaging diverse audiences in meaningful ways, ultimately motivating collective action toward addressing climate challenges.

Social Complexities. Focusing on ethics, equity, and justice, Rolnick et al. [42] emphasise that climate-related data often suffers from scarcity, incompleteness, or bias, making rigorous preprocessing and validation essential for ensuring reliability. Moreover, understanding the reasoning behind ML model predictions is crucial for real-world applications, particularly in high-stakes areas such as climate policy. To fully harness the potential of AI, it is vital to address disparities in its accessibility and deployment. Similarly, recent work [10] emphasises the importance of prioritising climate justice in the use of AI for climate action by ensuring that marginalised communities and regions most affected by climate change are not left behind.

4 Research Agenda: Mindful AI for Climate Policy

The tensions outlined in the previous section highlight the need for careful stewardship of powerful technologies like AI [3, 43] to ensure they contribute positively to climate goals rather than causing harm. Building on these insights, we propose a research agenda that explores the opportunities, challenges, trade-offs, and ethical dilemmas that arise as AI takes on a growing role in climate change policymaking. While our agenda is informed by the European policymaking context, the issues we address are broadly relevant to global climate governance. Many of these tensions transcend regional boundaries, emerging across diverse political and environmental landscapes.

4.1 Accessibility to climate data

Understandings of climate change rely heavily on data—metrics like atmospheric CO₂, global temperatures, and emissions shape perceptions of its impacts [34, 44, 45]. Yet such data is typically inaccessible to stakeholders outside of the scientific research domain due to factors such as ownership, interoperability, and illegibility [23]. Recent HCI work has highlighted the pressing need to improve the accessibility and salience of climate change data and models while broadening the scope of who is considered a user of environmental data [35].

Opportunity: Past work has called on HCI scholars to design systems that help policymakers interpret environmental data effectively [6] and illustrated the potential for AI to assist in converting atmospheric data into formats and metrics that are meaningful and understandable by different stakeholders [8]. Expanding this, our workshop highlighted the importance of taking a step back to carefully assess what data and metrics are truly needed and usable in specific contexts and how they can be contextualised with existing knowledge and insights. AI should be deployed selectively in policymaking contexts, recognising that some challenges may be best addressed through existing methodologies rather than novel technological solutions.

Trade-off: Improving data legibility for new stakeholders is often achieved by reducing complexity, which means that some data are included and other data excluded. These choices, whether conscious or unconscious, shape proposed solutions and policies [46]. Simplification might also lead to overlooking established climate science knowledge, reinforcing misconceptions or misinterpretations. Moreover, the social and environmental costs of deploying AI systems such as data transparency [42], energy consumption [12, 26], and resource depletion [18, 48] must be carefully weighed against their potential benefits.

Ethical Dilemma: The control of AI models and training data plays a crucial role in shaping future scenarios. AI models for climate forecasting and impact analysis are often developed and owned by private companies, meaning policymakers may have to rely on "black box" models without understanding their limitations or biases [41]. If only a few organizations have access to key data sources, foresight outcomes may reflect their interests rather than broader societal needs. This opacity raises concerns about accountability and trust in decision-making processes [21]. Furthermore, technology's unintended/undesirable consequences must be considered [40]. AI-driven solutions might introduce new dependencies, centralizing control over climate data interpretation and marginalizing community-driven environmental knowledge [16, 42].

To address these challenges, we call on the HCI community to design transparent, user-centred tools and frameworks that prioritize accessibility, inclusivity, and interpretability of climate data. These tools should empower diverse stakeholders to engage with, critique, and make informed decisions based on data, while ensuring ethical oversight of AI-driven methodologies. Employing research methods such as ethnographic studies and participatory design would

facilitate investigations into how different communities currently interact with climate data and to engage diverse stakeholders such as policymakers, scientists, and local businesses in co-designing climate data tools that reflect their needs and concerns.

We encourage HCI researchers to focus on designing modular and open-source systems (see for example, [5]) which will enable different stakeholders to customize and integrate climate data into their workflows, and explore multi-modal data representations such as visualizations, narratives, and interactive tools to suit varying governance sectors and expertise levels. Additionally, designers should prioritise low-carbon energy-efficient AI models and infrastructures to minimize the environmental footprint of computational processes (see for example, [9]). These strategies can create accessible, sustainable, and effective tools for engaging with climate data.

4.2 Supporting practical on-the-ground climate action

To support practical on-the-ground action, AI systems such as the policy prediction service explored in our workshop would need to integrate insights from a range of different sources, as perspectives that qualify as legitimate evidence are key for climate action [47, 51]. Aligning with past work that articulates scientific knowledge as politics [14] highlights the potential of data in “making things matter” [35], this agenda point centres around the role AI might play in making data actionable.

Opportunity: AI could potentially assist in contextualizing atmospheric data in relation to pre-existing knowledge (e.g., data on citizen perspectives and behaviours), providing holistic insights into both the impact of past initiatives and predictions of possible future impacts. However, past work has emphasised how the contextualization of climate data is not merely a technical task but is deeply intertwined with political and economic considerations [39]. It is important to consider how factors such as funding constraints, stakeholder interests, and governance structures play pivotal roles in determining how climate information is produced, interpreted and applied.

Challenge: Cities are complex, consisting of numerous interrelated but often siloed sectors (e.g., waste management, transport) where data is collected through different non-standardized methods [36]. Furthermore, existing knowledge from local communities and practitioners is often overlooked when designing AI-driven solutions. Ensuring that AI integrates with and enhances human expertise rather than disregarding it is critical.

Ethical Dilemma: Recent work by Lampinen and colleagues [28] has drawn attention to the considerable amount of human labour involved in the development and maintenance of AI and algorithmic systems—an aspect often ignored in discussions around potential applications of AI. For instance, annotating different datasets and stress-testing systems to identify vulnerabilities requires significant time and effort. Additionally, there is a risk that the reliance on AI-driven governance tools may reinforce existing power imbalances by centralizing decision-making and marginalizing community-led

climate action [41]. Reconciling these concerns with AI’s potential to support urban climate action necessitates ethical oversight and participatory development approaches.

To address these challenges, we call on the HCI community to adopt collaborative, interdisciplinary approaches focused on data standardization, transparency in AI development, and recognition of the human labour involved. AI tools for climate action should be rooted in the lived experiences and specific needs of local communities. Community-Based participatory research that involves local stakeholders in the AI design process (see for instance, [49]) could help to ensure that solutions are contextually relevant. Following this, mixed-methods evaluation processes that combine quantitative impact assessments with qualitative feedback from policymakers and citizens can be used to evaluate the effectiveness of AI-driven interventions in locally specific contexts.

We encourage HCI researchers to consider human-in-the-loop AI [2, 37] as a key design principle to ensure that AI systems complement, rather than replace, expert judgment in climate decision-making. Moreover, focus should be directed towards designing localized but scalable solutions for adaptable AI models that consider regional variations in climate governance. These strategies can help create AI systems that are transparent, effective, and tailored to the unique needs of cities.

4.3 Citizen participation in policymaking

Understanding how digital technologies support or hinder citizen participation in urban policymaking is a widely investigated topic within the HCI research community [24, 34, 47, 50, 53] but the potential role of AI is under-explored.

Opportunity 1: AI has the potential to assist in imagining alternative, climate-adapted lifestyles so that the public can familiarize themselves with required lifestyle changes, understand how these changes might manifest personally, and discuss diverse options [1, 15, 31]. However, it is crucial to ensure that these AI-generated insights build on existing environmental and behavioural knowledge rather than oversimplifying complex social transformations.

Opportunity 2: AI could help calculate the carbon cost of goods and services using available data and convert this into tailored forms of information for different stakeholders. For example, when shopping, AI could provide comparative emission information for similar items, supporting more sustainable consumer choices. However, the unintended consequences of such interventions—such as data biases or behavioural nudging without full transparency—should be considered [16, 40].

Trade-off: A key consideration is how the widespread application of such tools shapes the public’s understanding and acceptance of AI technologies [19], including awareness of their associated energy and carbon costs [18, 48]. While AI-driven platforms may enhance public engagement with sustainability, they also risk obscuring the environmental burden of AI itself.

Ethical Dilemma: Echoing past work on the effects of e-governance

on promoting active citizenship [33], delegating democratic participation to AI-based systems might marginalize certain groups, reduce transparency, and weaken accountability in decision-making. This raises concerns about whether AI-enabled participation is truly inclusive or whether it inadvertently reinforces digital divides.

To address the complexities of AI in climate change policy development, we urge the HCI community to consider the unintended consequences of AI-mediated public engagement. By applying participatory and co-design methodologies, we can promote transparency, inclusivity, and critical dialogue about the long-term social and environmental impacts of these technologies (see for instance, [22]). Ethnographic studies in HCI [25, 27] are also highly effective for understanding how AI-enabled participation is perceived and adopted by various stakeholders in climate governance, providing valuable insights into real-world challenges.

Practical design approaches must prioritize explainable AI, ensuring that AI-generated insights in policymaking are understandable and contestable (see for example, [4]). Fairness and inclusivity should guide AI platforms to reduce digital exclusion and power imbalances, ensuring equitable participation [10]. Additionally, sustainability in design is critical, promoting AI solutions that minimize energy consumption and consider environmental trade-offs [9]. By incorporating these principles, we can foster an AI-driven approach to climate policy that is socially responsible, inclusive, and sustainable.

4.4 Navigating Power Imbalances

AI-driven climate policymaking is shaped by power imbalances, as corporations, governments, and well-resourced nations control access to AI technologies and data [10]. This dominance risks sidelining marginalized communities, reinforcing global inequalities, and overlooking local environmental knowledge. Biases in AI systems can further entrench these disparities, leading to policies that favour dominant narratives over inclusive, context-sensitive solutions.

Challenge: AI's use in policymaking poses challenges such as unclear accountability, biases in training data, and insufficient attention to local contexts [12]. When flawed insights inform policy decisions, assigning responsibility for negative outcomes becomes difficult [21]. These issues are compounded when environmental costs of deploying AI systems are overlooked, especially in regions where resources are scarce, and the consequences of AI deployment may have unforeseen long-term effects.

Ethical dilemma: AI-supported policy development has the potential to exacerbate global inequalities, as wealthier nations often dominate access to advanced technologies, while poorer nations disproportionately suffer the impacts of climate change [10]. Furthermore, this technological divide risks overlooking existing knowledge from marginalized communities and failing to integrate their insights into policy solutions, which could undermine the effectiveness of AI-driven climate strategies [39]. Moreover, environmental costs of AI systems—such as e-waste—can and resource further deepen these disparities by imposing an undue burden on

less-resourced regions [18, 48].

Opportunity: Reflecting critically on questions of ownership, power, and profit in relation to AI-generated climate insights can reveal new perspectives and alternative pathways for action [41]. By considering the environmental cost of deploying AI systems, we can reconcile the technology's potential with its impact on the planet [42]. There is an opportunity to design AI systems that incorporate the best of both technological advancements and existing environmental knowledge, ensuring that their deployment does not exacerbate climate change but rather supports its mitigation.

To navigate the challenges of AI in climate solutions, we urge the HCI research community to use participatory methodologies that prioritize shared responsibility, diverse user needs, and societal impacts, rather than focusing only on technical efficiency. Participatory Action Research (PAR) (see for instance, [20]) approaches involve community members in every step of the research process, ensuring that those impacted by climate issues have a voice in AI design. While participatory futures design (see for instance, [7]) enables participants to explore different future scenarios and the role of AI, empowering communities to shape the direction of technology in climate policy.

In terms of practical applications, we encourage designers to focus on distributed and decentralized AI models [38], which give local communities control over their data, promote fairness, and reduce environmental costs. Additionally, data sovereignty and ethical practices are essential, ensuring data is ethically sourced and communities retain control over its use, prioritizing privacy and security [17]. These approaches can guide the development of AI systems that are inclusive, sustainable, and ethically grounded.

5 Conclusion

This paper has explored the complex opportunities and tensions involved in integrating AI into climate change policymaking. Drawing inspiration from ongoing co-design studies with interdisciplinary city stakeholders, our work aims to spark dialogue within the HCI community on how to balance AI's environmental costs with its potential to support climate action. To advance this conversation, we have proposed a research agenda centred on four priority areas: improving data accessibility, supporting practical climate action, enhancing citizen participation, and addressing power imbalances.

While AI can amplify climate action, HCI researchers must critically examine its limitations, biases, and environmental costs to ensure it complements, rather than replaces, existing knowledge and human agency. Although our research agenda is rooted in European policy contexts, the challenges and opportunities it addresses are globally relevant. The demand for transparent, context-sensitive, and equitable AI-driven climate policies extends across governance systems, underscoring the importance of interdisciplinary and participatory approaches.

We call on the HCI community to embrace these methodologies in designing AI systems that are not only transparent and equitable but also responsive to diverse contexts. By doing so, we can ensure that AI contributes responsibly and effectively to the urgent task of climate change mitigation and adaptation.

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